RefLens: Evidence-Grounded Citation Verification through LLM-Based Multi-Agent Framework

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Abstract

Accurate citation is crucial for academic research, yet 11-4% of papers contain reference errors, undermining research credibility. Existing tools rely on derived information like summaries, posing hallucination risks and failing to provide end-to-end verification. We introduce RefLens, a multi-agent framework that automates citation verification. RefLens offers two key innovations: (1) automating the entire workflow from PDF parsing to report generation, and (2) providing evidence-grounded verification where all decisions are based on direct quotes from original source documents. In a 35-participant user study, RefLens received an average rating of over 4.15 out of 5 across value, trust, and usability. RefLens is the first end-to-end automated system for comprehensive citation verification, offering practical utility for academic workflows.

Project Page — https://hoohahabighead.github.io/RefLens

Introduction & Related Work

The reliability of academic research is founded on accurate citation, yet 11-41% of published papers contain reference errors, severely undermining research reproducibility (Zhang and Abernethy 2024). The recent advancement of Large Language Models (LLMs) has exacerbated this issue by introducing 'citation hallucination'—the generation of plausible but non-existent sources—which further burdens already overloaded reviewers (Drozdz and Ladomery 2024).

While the field has evolved towards semantic verification of scientific claims (Wadden et al. 2020), existing interactive systems still primarily focus on bibliographic management or summary-based analysis, leaving users vulnerable to hallucination risks (Chelli et al. 2024). To date, a true end-to-end automated system that performs evidence-based semantic verification by directly acquiring original source documents has been absent. To overcome these limitations, this research proposes **RefLens**, a multi-agent LLM framework that performs verification based on direct quotes from source documents. RefLens aims to enhance the transparency and reliability of scholarly arguments by automating the entire verification process and fundamentally eliminating the hallucination risks inherent in summarized information.

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Figure 1: Example of a citation verification card in RefLens. [A] shows the target citation sentence. [B] displays the three-axis evaluation (Intent, Validity, Importance) with justifications. [C] presents the source paper's summary and the Core Evidence. The sidebar [D] shows the overall citation statistics dashboard for the entire paper.

Method: RefLens

RefLens automates the entire citation verification process through a modular pipeline architecture, adopting principles from recent surveys on LLM-based autonomous agents (Wang et al. 2023), as depicted in Figure 2, where multiple autonomous agents collaborate. The workflow is composed of a preprocessing phase to acquire and structure documents, followed by a hierarchical verification phase for analysis, culminating in an interactive report.

Preprocessing: Document Acquisition. This initial phase builds the foundational data for verification. A **PDF Processor (PP)** agent analyzes the target paper's layout, converting the PDF into structured text while precisely extracting in-text citations and the bibliography. Leveraging this bibliographic information (e.g., DOI, arXiv ID), a **Crawler (CR)** agent then automatically downloads the original source PDFs from online academic databases, creating a comprehensive Evidence Database.

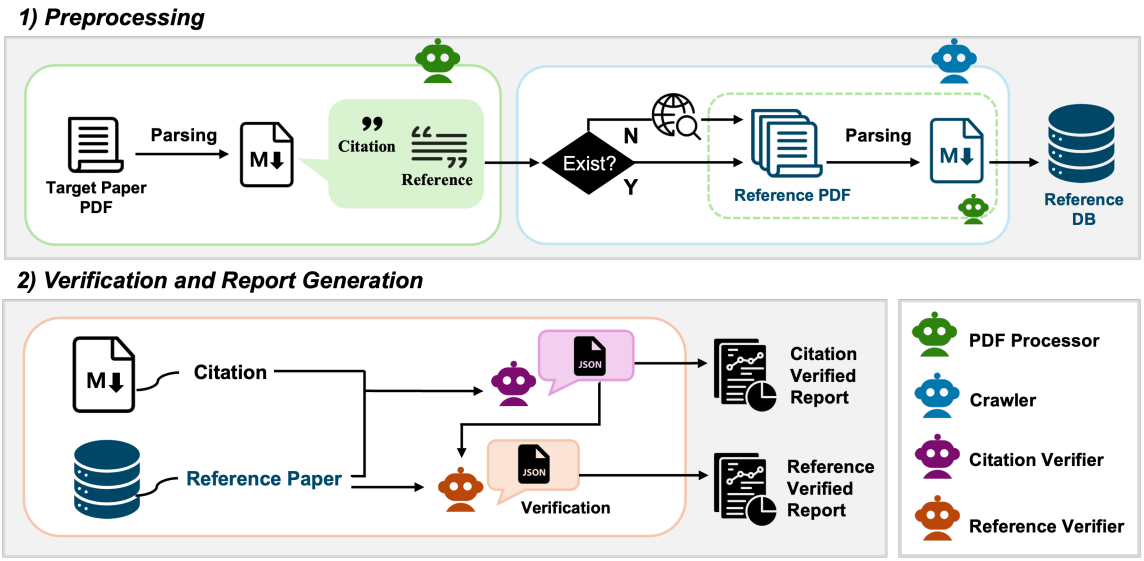

Figure 2: `RefLens` system pipeline overview. The framework operates through two main phases: (1) **Preprocessing** where the PDF Processor extracts citations and references from the target paper, and the Crawler automatically acquires missing reference documents to build a comprehensive database, and (2) **Verification and Report Generation** where the Citation Verifier and Reference Verifier analyze citation-source pairs to produce dual complementary reports for comprehensive citation validation.

Hierarchical Verification: Evidence-Grounded Analysis. This is the core phase where verification is performed using the constructed Evidence Database. The **Citation Verifier (CV)** agent conducts a microscopic analysis for each citation-reference pair based on a three-axis evaluation model: **Intent** (Cohan et al. 2019), **Validity** (Zhang and Abernethy 2024), and user-defined **Importance**. The key technology is a **Prompt-based Evidence Extraction** methodology. RefLens does not use common RAG techniques or vector embeddings. Instead, it directly leverages the up to 100M token context length of SOTA LLM models. It provides the model with the entire structured text of the original reference and the target citation in a single prompt, instructing it to identify and extract verbatim the most relevant text spans that support or contradict the claim. This approach ensures all verification decisions are anchored in the actual source text, fundamentally preventing hallucination risks. If extracted evidence is invalid or non-existent, it is added as a negative case example, incorporating a feedback loop that enhances accuracy. The results of this analysis are presented to the user in an interactive verification card, as shown in Figure 1. Finally, a **Reference Verifier (RV)** agent aggregates these micro-level results to perform a macroscopic, reference-level analysis and generate a strategic reading guide for the final report.

User Studies & Evaluations

To evaluate the practical utility of RefLens, we conducted a user study with 35 participants from diverse academic backgrounds (students, researchers). After experiencing a system demo, participants rated the system on a 5-point Likert scale across four dimensions. As shown in Table 1, results confirmed the system's high practical utility. Users rated its overall value and impact (4 questions, M=4.34), trustworthiness (4 questions, M=4.15), and usability (4 questions, M=4.19) highly. Strong adoption intention (3 questions, M=4.28) further validates that our evidence-grounded approach effectively addresses real-world research challenges. Details are in our Project Page.

Table 1: User evaluation results for key dimensions (N=35).

Dimension	Mean (SD)	Cronbach's α
Overall Value and Impact	4.34 (0.84)	0.89
Information Quality & Trust	4.15 (0.80)	0.82
System Usability	4.19 (0.77)	0.85
Adoption Intention	4.28 (0.89)	0.87

Conclusions & Future Work

This paper introduced RefLens, the first end-to-end, evidence-grounded framework for automated citation verification. We presented a novel methodology that utilizes prompt engineering instead of vector embeddings to extract evidence directly from source documents, and a user study with 35 participants demonstrated its strong utility and trustworthiness. This work not only provides a practical tool but also suggests a new paradigm for human-AI collaboration in ensuring scholarly integrity. By grounding LLM-generated analysis in verifiable evidence, RefLens demonstrates a path toward building more trustworthy and accountable AI systems for critical domains. Future work will proceed toward broader accuracy benchmarking and multilingual support. The system demo is available on our project page.

Acknowledgments

Add Ack

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